

## Statistics 8330 Final Exam – Fall 2023

You may use R or Python for all problems. Submit a single R and/or Python file with problems well delineated. You do NOT need to make an R Markdown or Jupyter notebook file – just the script file is fine. However, in order to full credit, your code must include comments that explain what you are doing. You may use all sources of information EXCEPT 1) direct communication with other people and 2) AI tools like ChatGPT. There will be partial credit for successfully doing portions of problems, even if you don't do everything. Upload your completed exam to the Final Exam drop box on eLC. Each problem is worth equal points. Good luck and happy holidays!

1. In this problem, you will conduct a small simulation study of how prediction error in the lasso depends on the penalty parameter. You should devise a small number of scenarios that vary in how they favor larger or smaller values of the penalty parameter. You then will conduct simulations that produce plots of mean prediction error versus the penalty parameter. In each simulation replicate, the model should be fit on training data and prediction error calculated for test data. Include comments that explain your simulation scenarios and briefly explain your findings.
2. Write a function that uses leave-one-out cross validation to find the optimal smoothing parameter for a kernel smoother. This function should take input of 1) a data frame, 2) the name of the kernel smoothing function, and 3) a range of values to test for the smoothing parameter. The output should be the optimal value of the smoothing parameter value. Use the NadWat function defined in class to test with.
3. Consider the following: UGA has five final exam days, each with three final exam periods. Suppose that a student has three final exams. Assume that the exams have an equal chance of falling in any of the 15 possible exam slots, but there can't be two exams at the same time (i.e. exam slots are sampled without replacement). What is the probability that the student's last final exam will be on each day? In this problem, you will write a function that finds this probability **by simulation** for a more generalized situation. Specifically, you will write a function of form

**exam\_day\_distribution(Days,nSlots,nExams,nrep)**

**Days** is a vector of the days on which exams occur, **nSlots** is the number of exam slots per day, **nExams** is the number of exams that the student has, and **nrep** is the number of simulation replicates (which should have a default value of 1000). It should output the

probability for the day that the student finishes their exams as shown below. You should include at least two examples of running your function. Note that your function must find the probabilities by simulation and not any direct calculation method.

**Hint: My code works by 1) indexing the possible exam slots from 1 to the total number, 2) sampling with replacement from those values, 3) finding the highest value among those values, 4) converting to an index for the day by dividing by the number of slots per day and rounding. You do not have to do it exactly this way, but this is one way that works.**

```
> final_exam_day(c("Wednesday","Thursday","Friday","Monday","Tuesday"),3,3,100000)
```

```
Wednesday Thursday Friday Monday Tuesday  
0.00240 0.04196 0.14120 0.29648 0.51796
```

